THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET C

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The prime factorisation of 5005 is: 1

(a) $7 \times 11 \times 13$ **(b)** $5 \times 7 \times 11 \times 13$ **(c)** $5 \times 7 \times 13$ **(d)** $5 \times 11 \times 13$

2. If $\text{HCF}(26, 169) = 13$, then $\text{LCM}(26, 169)$ is: 1

(a) 169 **(b)** 26 **(c)** 338 **(d)** 13

3. The HCF of two consecutive natural numbers is: 1

(a) their product **(b)** 1

(c) the smaller number **(d)** 0

4. A quadratic polynomial whose sum and product of zeroes are -3 and 2 respectively is: 1

(a) $x^2 - 2x + 3$ **(b)** $x^2 + 3x + 2$ **(c)** $x^2 - 3x + 2$ **(d)** $x^2 + 2x - 3$

5. The degree of the polynomial $4x^4 + 0x^3 + 0x^5 + 5x + 7$ is: 1

(a) 3 **(b)** 7 **(c)** 5 **(d)** 4

6. A student states: "The zeroes of $x^2 + 7x + 10$ are 2 and 5." The correct response is: 1

(a) Correct **(b)** Incorrect; zeroes are -2 and -5

(c) Only 5 is correct **(d)** Only 2 is correct

7. The pair of equations $x + 2y = 3$ and $2x + 4y = 6$ has: 1

(a) infinitely many solutions **(b)** exactly two solutions

(c) no solution **(d)** a unique solution

8. A student states: "If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, the pair of equations is inconsistent." The statement is: 1

(a) Correct

(b) Correct only if $c_1 = 0$

(c) Incorrect — the pair has infinitely many solutions and is consistent

(d) Cannot be determined

9. A student states: "Two lines that never intersect represent a consistent pair." The correct response is: 1

(a) Correct

(b) Correct only for $x = 0$

(c) Incorrect — parallel lines give no solution, so the pair is inconsistent

(d) Cannot be determined

10. The values of k for which $x^2 + kx + 9 = 0$ has equal roots are: 1

(a) ± 3 **(b)** ± 9 **(c)** ± 6 **(d)** 6 only

11. The nature of the roots of $x^2 + x + 1 = 0$ is: 1

(a) more than two real roots **(b)** two distinct real roots

(c) no real roots **(d)** two equal real roots

12. For $x^2 + x - 6 = 0$, Student A gets roots $2, -3$ and Student B gets $-2, 3$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Find the HCF and LCM of 336 and 54, and verify that HCF $\times$ LCM $=$ product of the numbers. **2**

14. A student says that the polynomial $x^2 + 1$ has two real zeroes. Justify or refute. **2**

15. Find the value of k for which the pair $x + 2y = 5$ and $3x + ky + 15 = 0$ has no solution. **2**

16. Solve by the elimination method: $x + y = 5$ and $2x - 3y = 4$. **2**

17. Solve by factorisation: $100x^2 - 20x + 1 = 0$. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Show that $11 \times 13 \times 17 + 17$ and $3 \times 5 \times 7 \times 11 + 11$ are composite numbers. **3**

19. If α and β are the zeroes of $2x^2 - 5x + 3$, find **3**

(i) $\frac{1}{\alpha} + \frac{1}{\beta}$

(ii) $\alpha^2\beta + \alpha\beta^2$

(iii) $(\alpha - \beta)^2$.

20. Of two supplementary angles, the larger exceeds the smaller by 18°. Find the angles. **3**

21. A student claims that a rectangle with perimeter 28 m and area 50 m² can be made. Justify or refute. **3**

22. Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If they travel in the same direction, they meet in 5 hours; if they travel towards each other, they meet in 1 hour. Find the speeds of the two cars. **3**

OR

A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid ₹27 for a book kept for seven days, while Susy paid ₹21 for a book kept for five days. Find the fixed charge and the charge for each extra day. **3**

23. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round, while Ravi takes 12 minutes. If they start at the same point at the same time in the same direction, after how many minutes will they meet again at the starting point? **3**

OR

Prove that $3\sqrt{2}$ is irrational. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. Find the zeroes of $3x^2 - x - 4$ and verify the relationship between the zeroes and coefficients. Also find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $3x^2 - x - 4$. **5**

OR

Case study: A garden arch is in the shape of a parabola given by $y = -x^2 + 6x - 5$ (in metres), where $y = 0$ is ground level. **5**

(i) Find the points where the arch meets the ground.

(ii) Find the width of the arch at the ground.

(iii) Verify the sum and product of the zeroes.

(iv) Find the height of the arch at $x = 3$.

25. A pole has to be erected at a point on the boundary of a circular park of diameter 13 m such that the difference of its distances from two diametrically opposite fixed gates A and B is 7 m. Is it possible? If yes, at what distances from the two gates should it be erected? **5**

OR

Case study: A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in ₹) was found to be 55 minus the number of toys produced that day. On a particular day the total cost of production was ₹750. **5**

(i) Form the quadratic equation.

(ii) Find the number of toys produced that day.

(iii) A student says only 25 toys could have been made. Comment.

END OF QUESTION PAPER — **Total: 50 Marks**